NeuroLink 3.0 — AIN3002 Deep Learning Course Project

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Project Overview

NeuroLink 3.0 is a privacy-preserving N-of-1 deep autoencoder for detecting longitudinal behavioral drift in keystroke dynamics. The system learns a personalised baseline from a single individual's own typing history and flags sessions that deviate from it using reconstruction error — without relying on any population norms.

Validation R^2 = 0.9586 · Test R^2 = 0.9604
ε = 7.9983 at δ = 10■■ (Differential Privacy via DP-SGD)
42,501 typing lessons · 17 months · keybr.com
Architecture: 12→32→16→10→16→32→12 (2,230 parameters)

Submission Structure

README.pdf — This file — project overview and file descriptions

code/

- **NeuroLink_3_0_v5_1_FINAL.ipynb** — Main notebook with all outputs pre-computed

report/

- **NeuroLink_3_0_FINAL.pdf** — Final paper — 7 pages, CVPR format
- **latex_source/** — Full LaTeX source (main.tex, main.bib, sec/*.tex, styles

poster/

- **NeuroLink_3_0_Poster.pdf** — A0 presentation poster

Notebook Overview (outputs pre-computed — no re-execution needed)

1. **Data Loading & Preprocessing** — Loads typing-data.json, parses timestamps, computes WPM
2. **Feature Engineering** — Computes all 12 features including residual EV and ESRP
3. **Architecture** — Defines undercomplete autoencoder 12→32→16→10→16→32→12
4. **DP Training** — Opacus DP-SGD, ε=8.0 target, σ=0.5994, C=1.0, best epoch=54
5. **Evaluation** — Per-feature R^2, variance-weighted R^2, ablation v4/v5/v5.1
6. **LDI & Anomaly Detection** — Bootstrap CI thresholds, 3-tier classification (HIGH/MARGINAL/NOR
7. **Ghost V3 Predictor** — Empirical drift-adjusted WPM predictor
8. **Dashboard** — 8-panel visualisation over 435 deployment days

Key Technical Contributions

1. Residual EV/ESRP

Reformulation from rolling-mean to temporal deviation resolves a structural reconstruction ceiling.
EV R^2: 0.234 → 0.954 ESRP R^2: 0.113 → 0.854

2. N-of-1 Design

Model trained exclusively on single-user data — no population norms required.